

Ahmad Fahim Basheer Ahmad

ahmadfahim2k@gmail.com | +44 7818436241 | Liverpool, United Kingdom | [linkedin.com/in/ahmadfahim2k](https://www.linkedin.com/in/ahmadfahim2k)

github.com/ahmadfahim2k | portfolio-self-pi-31.vercel.app

Data Scientist with hands-on experience in NLP, deep learning, RAG pipelines, and end-to-end ML systems, currently completing a Master's in Advanced Data Science and AI at the University of Liverpool. Backed by 3+ years as a full-stack developer in the healthcare and pharmaceutical domain, bringing strong software engineering practices to ML pipeline development and deployment

Education

Master of Advanced Data Science and Artificial Intelligence

University of Liverpool

Sep 2025 – Sep 2026 | Liverpool, United Kingdom

- Engineered disease-spreading simulation in C and implemented parallelization and optimization to achieve 28x speedup with OpenMP and 31x speedup with MPI
- Collaborated on a group project to build a full-stack financial crisis analysis tool, analysing 3,158 US equity tickers across crisis periods using Next.js, FastAPI, and LLM-generated insights
- Designed and analyzed algorithms for sorting and optimization problems, including pancake-sorting strategies and Hamming-code based encoding schemes

Bachelor of Technology in Computer Science and Engineering

B.S. Abdur Rahman Crescent Institute of Science and Technology

Jul 2018 – May 2022 | Chennai, India

- First Class with Distinction
- CGPA 8.73/10

Experience

Virtusa Consulting Services

Associate Engineer

Jun 2022 – Aug 2025 | Chennai, India

- Migrated three healthcare front-ends to a custom library, completing the project in 12 months
- Built reusable React components with Tailwind CSS and Recoil in Storybook, achieving 100% UI parity post-migration
- Fixed over 30 bugs post-migration, collaborating with QA and earning Star Performer of the Month
- Worked within Agile/Scrum teams, using JIRA to manage sprints, track deliverables, and coordinate cross-functional work
- Advanced from junior developer in a two-person team to sole developer on a dedicated project, owning requirements gathering, feasibility analysis, feature demos, and sprint delivery
- Conducted technical interviews for prospective hires and mentored new joiners in React to ensure readiness for client-facing project assignments

Projects

Stock Analyzer : Financial Crisis Analysis Tool

Full-stack web application for analysing US equity performance across major historical financial crises

Feb 2026 – Jun 2026

- Developed a FastAPI backend serving OHLC data for 3,158 US equity tickers (S&P 500, NASDAQ, NYSE, Forbes 2000) spanning 1970–2022
- Computed financial metrics per request including annualised volatility, average log return, ROI ratio, and recovery duration using pandas and NumPy
- Integrated Groq LLM to generate natural language summaries of computed metrics for a stock and time period
- Built an interactive Next.js frontend with Recharts visualisations for comparing stock performance across the Dot-Com crash, 2008 Financial Crisis, and COVID-19
- Aggregated yearly performance across 11 GICS sectors and implemented PDF export of charts and metrics
- Deployed the full stack to Azure Container Apps with a GitHub Actions CI pipeline validating Docker builds on every push, catching a build failure before deployment
- github.com/ahmadfahim2k/stock-analyzer-group-project 

ShoeBot : E-Commerce Chatbot [↗](#)

LLM-based shoe shopping assistant that handles product search, FAQs through a conversational interface

Apr 2026

- Implemented semantic router using HuggingFace to classify user intent and dispatch queries to focused models
- Created a natural language to SQL pipeline using Groq to query a web-scraped Flipkart product catalog
- Integrated FAQ retrieval using RAG with ChromaDB and deployed the chatbot as a Streamlit application
- github.com/ahmadfahim2k/shoebot-genai [↗](#)

Vehicle Damage Prediction [↗](#)

Deep learning system that classifies vehicle damage from images into 6 categories

Jan 2026

- Trained a vehicle damage image classifier using transfer learning with ResNet50 in PyTorch, achieving ~81% validation accuracy across 6 damage categories
- Tuned hyperparameters with Optuna and applied data augmentation to improve model generalization
- Served the model via a Streamlit app and FastAPI REST API for real-time inference on uploaded vehicle images
- github.com/ahmadfahim2k/dl-project-vehicle-damage-detection [↗](#)

Credit Risk Modelling [↗](#)

ML-powered web app that predicts loan default risk and generates a credit score with rating

Sep 2025

- Built a credit risk prediction model using Logistic Regression with SMOTE and Optuna tuning, achieving ~93% accuracy on 50,000 loan records
- Engineered features loan-to-income and delinquency ratios, from merged customer, loan, credit bureau datasets
- Applied VIF and Information Value analysis to reduce multi-collinearity and select the most predictive features
- Converted default probability into a 300–900 credit score with a 4-tier rating (Poor/Average/Good/Excellent)
- Deployed an interactive Streamlit web app for real-time credit risk assessment
- github.com/ahmadfahim2k/ml-project-credit-risk-model [↗](#)

Healthcare Premium [↗](#)

Regression model that predicts annual health insurance premiums from user inputs

Aug 2025

- Developed an end-to-end ML pipeline with XGBoost to predict healthcare insurance premiums, achieving R^2 of 0.998 on unseen data
- Applied age-stratified modelling with separate models for young and older cohorts after error analysis revealed distinct pricing patterns
- Constructed a composite medical risk score from 4 chronic condition indicators to improve accuracy
- Launched a Streamlit app that routes user inputs to age-stratified models and returns premium estimates
- github.com/ahmadfahim2k/ml-project-healthcare-premium [↗](#)

Skills

Languages — Python, JavaScript, C, SQL

Frontend — React, Tailwind CSS, Recoil, Storybook

Backend — Node.js, Django, FastAPI

ML / AI — PyTorch, Scikit-Learn, XGBoost, LangChain, ChromaDB, HuggingFace, Optuna, RAG

Tools — Git, Streamlit, Docker, Azure

Practices — Agile, Scrum, JIRA, CI/CD

Certificates

• [CodeBasics: Deep Learning](#) [↗](#)

• [CodeBasics: Machine Learning](#) [↗](#)